5.3 Positive Classes and the Order

Section 5.2 made $\mathbb{R}$ a field: its elements are classes of Cauchy sequences of rationals, added and multiplied through representatives. A field is not yet a number line. To compare two real numbers we need an order, and it must cooperate with the arithmetic: $\mathbb{R}$ should be an ordered field in the sense of Definition 2.4.1. Chapter 3 ordered $\mathbb{Q}$ in two steps. It first said which rationals are positive, and then let $a < b$ mean that $b - a$ is positive (Proposition 2.4.1). This section takes the same two steps for $\mathbb{R}$. All the work is in the first: deciding which classes are positive, in a way that does not depend on the representative.

Notation is that of Sections 5.1 and 5.2: $\mathcal{C}$, $\sim$, $[a]$, the operations of Definition 5.2.2, $0_{\mathbb{R}} = [(0)_n]$, $1_{\mathbb{R}} = [(1)_n]$, and $[a] - [b] = [a - b]$, used without comment. Until $\mathbb{R}$ has an order, every margin, tolerance and threshold is rational or natural: we may not measure with what we are building.

5.3.1 Comparing terms does not compare classes

The first idea is to compare two classes by comparing their terms. It fails, and the way it fails shows what the definition must contain.

Example. Termwise Order Is Not an Order on Classes

Consider the rule

> $[a] < [b]$ means that $a_n < b_n$ for every n from some index on.

Let $a_n \coloneqq 0$ and $b_n \coloneqq 1/(n+1)$ for $n \in \mathbb{N}$. Show that $[a] = [b]$. Then apply the rule to the classes $[a]$ and $[a]$ twice: once through the representatives a and a, once through a and b. Conclude that the rule does not define a relation on $\mathbb{R}$.

Solution. Both sequences are in $\mathcal{C}$: a constant sequence and a null sequence are Cauchy (Theorem 4.5.3 (a)), and $(1/(n+1))$ is null in $\mathbb{Q}$ (Example (First Limits) (b)). Termwise, $b - a = (1/(n+1))$, which is null, so $b \sim a$, and $[a] = [b]$ by Sets and Functions, Theorem 5.2.1 (b).

Through the representatives a and a: the rule asks whether $0 < 0$ from some index on, and $0 < 0$ never holds. So the rule says that $[a] < [a]$ is false.

Through the representatives a and b, which name the same two classes, since $b \in [b] = [a]$: for **every** n, $a_n = 0 < 1/(n+1) = b_n$. So the rule says that $[a] < [a]$ is true.

A relation on $\mathbb{R}$ must give one answer for each pair of classes. The rule gives two answers for the pair $(0_{\mathbb{R}}, 0_{\mathbb{R}})$, depending on the representatives, so it defines no relation on $\mathbb{R}$. Worse, the second answer, $0_{\mathbb{R}} < 0_{\mathbb{R}}$, is excluded by trichotomy (O1) in any ordered field.

The trouble is that the gap between a_n and b_n is allowed to shrink to nothing, and a gap that shrinks to nothing is exactly what $\sim$ ignores. A class that deserves to be called positive must have terms that stay above 0 by a **fixed** margin.

5.3.2 Positive sequences

We look for a property of a single sequence $a \in \mathcal{C}$ that says "the number a names is greater than 0". The example shows that "eventually $a_n > 0$" is too weak, because a null sequence can have it. The repair is to name the margin first.

Positive means: eventually above one fixed positive rational.

Definition 5.3.1. Positive Cauchy Sequence

A sequence $a \in \mathcal{C}$ is **positive** if **there exist** $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ such that $a_n \geq \delta$ **for every** $n \in \mathbb{N}$ with $n \geq N$. In symbols,

$$\exists \delta \in \mathbb{Q}_{>0} \;\; \exists N \in \mathbb{N} \;\; \forall n \in \mathbb{N} \;\; (n \geq N \Rightarrow a_n \geq \delta).$$

In words: the first quantifier names a margin δ, a positive rational, before anything else is named; the second names a threshold; the third says that **every** term from the threshold on is at least the margin. Because δ comes first, one margin serves all the late terms at once. The word "positive" describes the sequence as a name for a number, not its terms one by one: the terms before N may be negative, and, as the warning below shows, terms that are all > 0 are not enough.

Compare the weaker condition

$$\exists N \in \mathbb{N} \;\; \forall n \in \mathbb{N} \;\; (n \geq N \Rightarrow a_n > 0).$$

There each late term clears 0 by its own amount, which may shrink with n; in the definition one margin is named before n, and that order of the quantifiers is the whole point.

Examples.

A positive constant. For $q \in \mathbb{Q}_{>0}$, the constant sequence $(q)_n$ is positive, with $\delta = q$ and $N = 0$. This degenerate case is how Section 5.4 will see that the rational q, placed inside $\mathbb{R}$, is greater than 0.

The Newton sequence. Let (x_n) be the sequence of Example (A Rational Sequence Aimed at the Gap), which is in $\mathcal{C}$ by Theorem 4.6.1. The same theorem states that $x_n > 1$ for every $n \in \mathbb{N}$. So (x_n) is positive, with $\delta = 1$ and $N = 0$.

Late positivity. Let $a_n := 1 - 2/(n+1)$. It is in $\mathcal{C}$, as the difference of a constant sequence and a constant multiple of the null sequence $(1/(n+1))$ (Theorem 4.5.3 (a), (b)). Its first terms are $-1, 0, 1/3, 1/2$, so it is not positive "from the start". For $n \geq 3$, $n+1 \geq 4$, so $1/(n+1) \leq 1/4$ (Proposition 4.1.2 (c)), hence $2/(n+1) \leq 2/4 = 1/2$ (Proposition 4.1.2 (d), with $0 \leq 2 \leq 2$), and $a_n \geq 1/2$. So a is positive, with $\delta = 1/2$ and $N = 3$. No smaller N works with this δ, since $a_2 = 1/3 < 1/2$; the definition allows any finite stretch of bad terms.

Non-example by a minimal change. Keep every term positive, but let the margin shrink: $a_n := 1/(n+1)$.

↻ **Second reading** · *Why this sequence is not positive*

Every term is > 0, so the weaker condition above holds with $N = 0$. What fails is the order of the quantifiers. The negation of the definition, by Logic and Proof, Proposition 6.2.3 and Logic

and Proof, Proposition 6.2.2, reads

$$\forall \delta \in \mathbb{Q}_{>0} \;\; \forall N \in \mathbb{N} \;\; \exists n \in \mathbb{N} \;\; (n \geq N \wedge a_n < \delta),$$

and it holds: given δ and N, Corollary 3.2.4 gives $n_0 \in \mathbb{Z}_{>0}$ with $1/m < \delta$ for every $m \geq n_0$. With $n := \max\{N, n_0\}$, we have $n \geq N$, and $m := n+1 \geq n_0$, so $a_n = 1/(n+1) < \delta$. Every margin is eventually undercut.

Why this definition. Each clause has a job.

- *The margin δ.* Without it, the definition would be the weaker condition, and Example (Termwise Order Is Not an Order on Classes) shows the result: $(1/(n+1))$ would be positive, while $(0)_n$, which names the same number $0_{\mathbb{R}}$, is not.
- *A rational margin.* The only positive numbers available are rationals. A real margin would use the order of $\mathbb{R}$, which is what we are defining.
- *The threshold N.* Changing finitely many terms does not change the class (Example (Finite Changes and Shifts)), so a definition meant for classes must ignore every finite stretch of terms. The late-positivity example needs $N = 3$.

↻ **Second reading** · *Why $\geq$ and not $>$*

Either inequality would do: $a_n \geq \delta$ gives $a_n > \delta/2$, and $a_n > \delta$ gives $a_n \geq \delta$. We use $\geq$, the form in which Theorem 4.5.4 delivers its conclusion.

Warning. "Every term of a is positive" does **not** make a positive. The sequence $(1/(n+1))$ has every term > 0, and it is not positive (non-example above); its class is $0_{\mathbb{R}}$.

Quick check. Let $a \in \mathcal{C}$, with δ ranging over $\mathbb{Q}_{>0}$. Which of the two conditions

$$\text{(A)} \;\; \exists \delta \; \exists N \; \forall n \geq N \;\; a_n \geq \delta, \qquad \text{(B)} \;\; \exists N \; \forall n \geq N \; \exists \delta \;\; a_n \geq \delta$$

implies the other? Which of them says the same as "eventually $a_n > 0$"?

Solution.

(A) implies (B): the margin found once serves as the δ for every late n. (B) does not imply (A): $(1/(n+1))$ satisfies (B), with $\delta := 1/(n+1)$ chosen after n, and fails (A). (B) says the same as "eventually $a_n > 0$": if $a_n > 0$, take $\delta := a_n$; if $a_n \geq \delta > 0$, then $a_n > 0$. So (A) is the definition of positive, and (B) is the weaker condition.

Payoff: positivity belongs to the class. A definition about sequences says something about a class only if equivalent sequences agree on it. The proof shows the method of this section in its simplest form.

Proposition 5.3.1. Positivity Is a Property of the Class

Let $a, a' \in \mathcal{C}$ with $a' \sim a$. If a is positive, then a' is positive.

Idea (Scratch work). The goal is an existence statement whose first quantifier is a margin. Name it first, from the data, before any index appears; then ask the given sequences for thresholds.

Givens	Goal
$a_n \geq \delta$ for $n \geq N$, with $\delta \in \mathbb{Q}_{>0}$; $a' - a$ null	$\exists \delta' \in \mathbb{Q}_{>0}\ \exists N'\ \forall n \geq N'\ \ a'_n \geq \delta'$
the margin comes first and may depend only on δ: afford to lose half, $\delta' \coloneqq \delta/2$	$\exists N'\ \forall n \geq N'\ \ a'_n \geq \delta/2$
nullity of $a' - a$ at the tolerance $\delta/2$: $a'_n - a_n$ is within $\delta/2$ of 0 for $n \geq N''$	the threshold $N' \coloneqq \max\{N, N''\}$
for $n \geq N'$: $a'_n > a_n - \delta/2 \geq \delta - \delta/2$	gap closed

Proof. Fix $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $a_n \geq \delta$ for every $n \geq N$. Then $\delta/2 \in \mathbb{Q}_{>0}$ (Proposition 4.1.2 (b)). Since $a' - a$ is null, fix N'' with $|a'_n - a_n| < \delta/2$ for every $n \geq N''$, and let $N' \coloneqq \max\{N, N''\}$. For $n \geq N'$, Proposition 4.1.3 (d) gives $-\delta/2 < a'_n - a_n$, so $a'_n > a_n - \delta/2 \geq \delta - \delta/2 = \delta/2$ (Proposition 2.4.3 (b), Proposition 4.1.2 (b)). So a' is positive, with margin $\delta/2$ and threshold N'. ∎

This is Proposition 4.5.5 with $F = \mathbb{Q}$, written out because its order of choices is the method of this section.

The margin halves, and that is harmless: the definition asks only for **some** margin. Exercise C2 shows a tempting proof of this proposition that keeps a margin of 0, and so does not prove it.

Example. Method: name the margin before the index

Template. To prove that $b \in \mathcal{C}$ is positive, or that $[a] < [b]$: (1) name a margin $\delta \in \mathbb{Q}_{>0}$ from data that do not depend on n: a given margin, or a fraction (such as half) of a given margin or of a positive value the terms visibly approach; (2) only then ask the given sequences (Cauchy, null, positive) for thresholds at tolerances computed from δ; (3) take the largest of these finitely many thresholds and check that the terms of the sequence to be shown positive (b, or $b - a$ for $[a] < [b]$) are $\geq \delta$ beyond it.

When to use it. Whenever the goal reads $\exists \delta\ \exists N\ \forall n \geq N\ \ldots$: positivity, a strict inequality between classes, a bound that must hold beyond one index.

The usual failure. Proving "$b_n > 0$ for all large n" and calling b_n the margin: it depends on n (the Quick check above; Exercise C2).

We can now name the set of positive real numbers. Let

$$P := \{[a] : a \in \mathcal{C} \text{ and } a \text{ is positive}\} \subseteq \mathbb{R}.$$

Chapters 2 and 3 built the positive sets $P_{\mathbb{Z}}$ and $P_{\mathbb{Q}}$; P is the same step for $\mathbb{R}$. For $x \in \mathbb{R}$:

$$x \in P \iff \text{some representative of } x \text{ is positive} \iff \text{every representative of } x \text{ is positive.}$$

The first equivalence is the definition of P. For the second, every class has a representative (Sets and Functions, Theorem 5.2.1 (a)), which gives "every $\Rightarrow$ some". Conversely, if $x = [a]$ with a positive and $b \in x$, then $b \sim a$ (Sets and Functions, Definition 5.2.2), so b is positive by Proposition 5.3.1. Membership in P can therefore be tested on **any** representative we like. This is the well-definedness the termwise rule lacked.

5.3.3 The positive classes

Chapter 2's recipe needs two facts about P: it is closed under sum and product, and it splits $\mathbb{R}$ into three parts, P, $\{0_{\mathbb{R}}\}$ and the negatives of P. These are clauses (P1) and (P2) of Proposition 2.4.1.

Theorem 5.3.2. The Positive Real Numbers

(i) For every $x \in \mathbb{R}$, **exactly one** of $x \in P$, $x = 0_{\mathbb{R}}$, $-x \in P$ holds.

(ii) If $x \in P$ and $y \in P$, then $x + y \in P$ and $xy \in P$.

Idea. For (i), "at least one" is the sign theorem of Chapter 4: a Cauchy sequence that is not null is eventually above c or eventually below $-c$ (Theorem 4.5.4). A null representative gives $0_{\mathbb{R}}$. "At most one" holds because a sequence cannot be eventually above a positive margin and also eventually below a negative one, or null. For (ii), add or multiply the two margins and take the later of the two thresholds.

Proof. Let $x \in \mathbb{R}$, and write $x = [a]$ with $a \in \mathcal{C}$.

(i), *at least one.* Suppose first that a is null. Termwise $a - (0)_n = a$, so $a \sim (0)_n$ and $x = [(0)_n] = 0_{\mathbb{R}}$. Suppose next that a is not null. By Theorem 4.5.4 with $F = \mathbb{Q}$, there are $c \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ such that either $a_n \geq c$ for every $n \geq N$, or $a_n \leq -c$ for every $n \geq N$; fix them. In the first case a is positive, so $x \in P$. In the second case $-a_n \geq c$ for every $n \geq N$ (Proposition 2.4.3 (a), in each case of $\leq$), so $-a$ is positive and $-x = [-a] \in P$.

(i), *at most one.* We exclude the three pairs.

- $x \in P$ and $x = 0_{\mathbb{R}}$. Then $(0)_n$ is a representative of x, so it is positive, since every representative of an element of P is: some $\delta \in \mathbb{Q}_{>0}$ has $0 \geq \delta$. This contradicts $\delta > 0$ by (O1) in $\mathbb{Q}$.
- $-x \in P$ and $x = 0_{\mathbb{R}}$. Then $-x = [-(0)_n] = [(0)_n] = 0_{\mathbb{R}}$, and the previous case, applied to $-x$, excludes this.
- $x \in P$ and $-x \in P$. Then a and $-a$ are both positive: $a_n \geq \delta$ for $n \geq N$ and $-a_n \geq \delta'$ for $n \geq N'$, with $\delta, \delta' \in \mathbb{Q}_{>0}$. At $n := \max\{N, N'\}$, $\delta \leq a_n \leq -\delta'$ (Proposition 2.4.3 (a), in each case of $\leq$), so $0 < \delta \leq -\delta' < 0$, against (O1) in $\mathbb{Q}$.

(ii) Let $x, y \in P$, and write $x = [a]$, $y = [b]$. Both a and b are positive: there are $\delta_a, \delta_b \in \mathbb{Q}_{>0}$ and $N_a, N_b \in \mathbb{N}$ with $a_n \geq \delta_a$ for $n \geq N_a$ and $b_n \geq \delta_b$ for $n \geq N_b$. Let

$N := \max\{N_a, N_b\}$ and $n \geq N$. Adding the two inequalities (Proposition 2.4.3 (b)),

$$a_n + b_n \geq \delta_a + \delta_b,$$

and $\delta_a + \delta_b > 0 + 0 = 0$ by the same rule. Multiplying, by Proposition 4.1.2 (d) applied to $0 \leq \delta_a \leq a_n$ and $0 \leq \delta_b \leq b_n$,

$$a_n b_n \geq \delta_a \delta_b,$$

and $\delta_a \delta_b > 0$ by (O4). So $a + b$ and ab are positive, with margins $\delta_a + \delta_b$ and $\delta_a \delta_b$ and the threshold N. Hence $x + y = [a + b] \in P$ and $xy = [ab] \in P$ (Definition 5.2.2).

■

5.3.4 The order of the reals

With P in hand the definition is the one Chapters 2 and 3 used for $\mathbb{Z}$ and $\mathbb{Q}$.

Definition 5.3.2. Order on the Reals

For $x, y \in \mathbb{R}$, $x < y$ means $y - x \in P$. As usual, $x \leq y$ means $x < y$ or $x = y$, and $y > x$ means $x < y$.

In terms of representatives: $[a] < [b]$ exactly when $b - a$ is positive, that is, when $b_n - a_n \geq \delta$ from some index on, for some margin $\delta \in \mathbb{Q}_{>0}$. The terms of b must eventually overtake those of a by a fixed rational amount. Since $0_{\mathbb{R}} - x = -x$ and $x - 0_{\mathbb{R}} = x$, we have $x > 0_{\mathbb{R}}$ if and only if $x \in P$, and $x < 0_{\mathbb{R}}$ if and only if $-x \in P$.

Theorem 5.3.3. The Reals Form an Ordered Field

With the operations of Definition 5.2.2 and the order of Definition 5.3.2, $\mathbb{R}$ is an ordered field (Definition 2.4.1), and its positive elements are exactly the elements of P.

Idea. Each order axiom is a statement about one difference, and Theorem 5.3.2 is exactly what Proposition 2.4.1 asks for. (O1) is (i) applied to $y - x$, since $-(y - x) = x - y$. (O2) is closure under sums, since $(z-y)+(y-x) = z-x$. (O3) holds because $(y+z)-(x+z) = y-x$. (O4) is closure under products.

Proof. By Theorem 5.2.3, $\mathbb{R}$ is a field, in particular a commutative ring (Definition 2.3.1). By Theorem 5.3.2, the subset $P \subseteq \mathbb{R}$ satisfies clause (P1) of Proposition 2.4.1 (part (ii)) and clause (P2) (part (i)). That proposition then says that the relation "$y - x \in P$", which is $x < y$ of Definition 5.3.2, makes $\mathbb{R}$ an ordered commutative ring with $P = \{x \in \mathbb{R} : x > 0_{\mathbb{R}}\}$. An ordered commutative ring that is a field is an ordered field (Definition 2.4.1).

■

So $\mathbb{R}$ is an ordered field, and every result that Chapters 2 and 4 proved for an arbitrary ordered field holds in it. We write $\mathbb{R}_{>0} := \{x \in \mathbb{R} : x > 0_{\mathbb{R}}\}$, the set $F_{>0}$ of Chapter 4 for $F = \mathbb{R}$; by the theorem, $\mathbb{R}_{>0} = P$. The rules for inequalities (Proposition 2.4.3), the rules for halves

(Proposition 4.1.2), the absolute value (Definition 4.1.1, Proposition 4.1.3), and limits, null sequences and Cauchy sequences with tolerances in $\mathbb{R}_{>0}$ (Definition 4.3.2, Definition 4.5.1) are now available in $\mathbb{R}$. From here on they may be applied with $F = \mathbb{R}$. It pays to name the field at each such use, so that a tolerance in $\mathbb{Q}_{>0}$ is never confused with one in $\mathbb{R}_{>0}$. One consequence deserves a line. In any ordered field, $x \leq y$ holds if and only if $y < x$ fails: by (O1), exactly one of $x < y$, $x = y$, $y < x$ holds.

Example. A Margin Read Off the Limit

Let $a_n := (2n+1)/(n+3)$. Show that $a \in \mathcal{C}$ and $[a] > 1_{\mathbb{R}}$, and find a margin and a threshold.

Solution. Since $2n+1 = 2(n+3)-5$, we have $a_n = 2-5/(n+3)$. The sequence $(1/(n+3))_n$ is the shift by 2 of the null sequence $(1/(n+1))_n$ (Proposition 4.3.2 (b), Example (First Limits) (b)), so a is a constant minus a multiple of a null sequence and lies in $\mathcal{C}$ (Theorem 4.5.3 (a), (b)).

By Definition 5.3.2 we need $a - (1)_n$ to be positive. Its terms are

$$a_n - 1 = \frac{n-2}{n+3} = 1 - \frac{5}{n+3}.$$

Margin first. These terms approach 1, so we aim at half of it, $\delta := 1/2$, before naming any index.

Then the index. Multiplying by $2(n+3) > 0$, or by its reciprocal (Proposition 2.4.3 (c), with equality as the other case),

$$\frac{n-2}{n+3} \geq \frac{1}{2} \iff 2(n-2) \geq n+3 \iff n \geq 7.$$

So $\delta = 1/2$ and $N = 7$ work, and $[a] > 1_{\mathbb{R}}$. No smaller threshold works with this margin, since $a_6 - 1 = 4/9 < 1/2$.

Any margin $\delta < 1$ works with a later threshold; $\delta = 1$ fails, since every $a_n - 1 = 1-5/(n+3) < 1$. The tempting shortcut, "$a_n - 1 > 0$ for $n \geq 3$", is the weaker condition and proves nothing about the class.

5.3.5 Comparing classes through their terms

The order is defined through margins, but in practice we usually know a termwise inequality and want an inequality between classes. The next lemma is the bridge, and the example at the start of the section shows its limit.

Lemma 5.3.4. Termwise Inequalities Pass to Classes

Let $a, b \in \mathcal{C}$ and $N_0 \in \mathbb{N}$. If $a_n \leq b_n$ for every $n \geq N_0$, then $[a] \leq [b]$.

Idea. Suppose instead that $[b] < [a]$. Then $a - b$ is positive, so from some index on a_n exceeds b_n by a fixed margin; one index beyond both thresholds then contradicts $a_n \leq b_n$. By trichotomy, "not $[b] < [a]$" is "$[a] \leq [b]$".

Proof. Suppose that $[b] < [a]$. By Definition 5.3.2, $[a] - [b] = [a - b] \in P$, so $a - b$ is positive, since every representative of an element of P is (Proposition 5.3.1). Fix $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $a_n - b_n \geq \delta$ for every $n \geq N$. Let $n := \max\{N, N_0\}$. Then $a_n - b_n \geq \delta > 0$. But $n \geq N_0$, so $a_n \leq b_n$, that is, $a_n - b_n \leq 0$ (Proposition 2.4.3 (a), in each case of $\leq$). These contradict (O1) in $\mathbb{Q}$. Hence $[b] < [a]$ is false. By (O1) in the ordered field $\mathbb{R}$ (Theorem 5.3.3), $[a] < [b]$ or $[a] = [b]$; that is, $[a] \leq [b]$.

■

Warning. The strict version of the lemma is **false**. In Example (Termwise Order Is Not an Order on Classes), $a_n = 0 < 1/(n+1) = b_n$ for **every** n, and yet $[a] = [b]$, so $[a] < [b]$ fails. A strict termwise inequality passes to classes only as $\leq$. To get $<$ between classes, show that the difference is eventually above a fixed margin.

Most uses of the lemma compare a class with a constant: if $a_n \leq q$ for every $n \geq N_0$, then $[a] \leq [(q)_n]$; and if $a_n \geq q$ for every $n \geq N_0$, then $[a] \geq [(q)_n]$.

Quick check. Let $a_n := 1/(n+1) - 1/(n+1)^2$. Every term satisfies $a_n \geq 0$. Is a positive?

Solution. No. Here $a_n = n/(n+1)^2$, so $0 \leq a_n \leq 1/(n+1)$ for every n (as $n \leq n+1$). The lower sequence $(0)_n$ and the upper sequence $(1/(n+1))$ are null, so a is null by the squeeze, Proposition 4.4.4 (c) with $F = \mathbb{Q}$; being null, $a \in \mathcal{C}$ (Theorem 4.5.3 (a)). Then $[a] = 0_{\mathbb{R}}$, as in the proof of Theorem 5.3.2 (i), and $0_{\mathbb{R}} \notin P$ by the same part. So a is not positive, although no term is negative. Lemma 5.3.4 gives only $[a] \geq 0_{\mathbb{R}}$, which is true.

Since $\mathbb{R}$ is an ordered field, every $x \in \mathbb{R}$ has an absolute value $|x|$ (Definition 4.1.1 with $F = \mathbb{R}$); Exercise C1 shows that it can be computed on representatives, $|[a]| = [(|a_n|)_n]$.

5.3.6 Exercises

5.3.6.1 A. Check your understanding

Exercise. A1

In parts (a)–(d), decide for each sequence whether it is in $\mathcal{C}$ and whether it is positive; when it is positive, give a margin δ and a threshold N.

(a) $a_n = 1 - 1/(n+1)$.

(b) $a_n = 1/(n+1)$.

(c) $a_k = d_k - 1$, where (d_k) is the sequence of decimal truncations of Example (The Truncations and the Newton Sequence Are Equivalent).

(d) $a_n = (-1)^n$.

(e) Show from Definition 5.3.2 and Definition 5.3.1 that $1_{\mathbb{R}} > 0_{\mathbb{R}}$. Which result of Chapter 2 gives the same conclusion in every ordered field?

(f) In the proof of Proposition 5.3.1, why is the tolerance for $a' - a$ taken to be $\delta/2$ and not δ? What would the proof give with δ?

Solution.

(a) In $\mathcal{C}$ (a constant minus a null sequence, Theorem 4.5.3 (a), (b)), and positive. For $n \geq 1$, $n+1 \geq 2$, so $1/(n+1) \leq 1/2$ (Proposition 4.1.2 (c)) and $a_n \geq 1/2$. So $\delta = 1/2$, $N = 1$ work. ($N = 0$ fails for this δ: $a_0 = 0$.)

(b) In $\mathcal{C}$, being null, and not positive: this is the non-example after Definition 5.3.1.

(c) In $\mathcal{C}$, as the difference of $(d_k) \in \mathcal{C}$ (Example (The Truncations and the Newton Sequence Are Equivalent)) and a constant, and positive. The first truncations are $d_0 = 1$ and $d_1 = 7/5$ (14 is the largest natural number whose square is at most $2 \cdot 10^2 = 200$), and (d_k) is non-decreasing ((T1), stated before Example (The Truncations and the Newton Sequence Are Equivalent)). So $d_k \geq d_1 = 7/5$ for every $k \geq 1$, and $a_k \geq 2/5$. So $\delta = 2/5$, $N = 1$ work. ($N = 0$ fails for every $\delta > 0$: $a_0 = 0$.)

(d) Not in $\mathcal{C}$: for every N, the indices N and $N+1$ give terms differing by 2, so the tolerance 1 has no threshold. The question "positive?" does not arise; the definition applies only to members of $\mathcal{C}$.

(e) $1_{\mathbb{R}} - 0_{\mathbb{R}} = [(1)_n - (0)_n] = [(1)_n]$, and $(1)_n$ is positive with $\delta = 1$, $N = 0$. So $1_{\mathbb{R}} - 0_{\mathbb{R}} \in P$, that is, $0_{\mathbb{R}} < 1_{\mathbb{R}}$. In every ordered commutative ring with $1 \neq 0$, Proposition 2.4.3 (e) gives $1 > 0$; for $\mathbb{R}$, $1_{\mathbb{R}} \neq 0_{\mathbb{R}}$ holds by Theorem 5.2.2.

(f) With the tolerance δ the proof gives only $a'_n > a_n - \delta \geq 0$: terms above 0 with no margin, which is the weaker condition. The tolerance $\delta/2$ leaves the margin $\delta/2$, named before n. Any tolerance strictly between 0 and δ would do.

5.3.6.2 B. Practice

Exercise. B1

Let $a_n = 1 - 1/(n+1)$ and $b_n = x_n - 1$, with (x_n) the Newton sequence of Example (A Rational Sequence Aimed at the Gap). Find margins and thresholds for a and b, and state the margins and threshold that the proof of Theorem 5.3.2 (ii) produces for $a + b$ and ab.

Hint: for b, compare x_n with $7/5$ by squaring.

Solution. The example. For a, Exercise A1 (a) gives $\delta_a = 1/2$, $N_a = 1$. For b: $x_n > 0$ and $x_n^2 > 2$ for every n (Example (A Rational Sequence Aimed at the Gap)), while $(7/5)^2 = 49/25 < 2$. Both $7/5$ and x_n are ≥ 0, so Proposition 2.4.3 (g) turns $(7/5)^2 < x_n^2$ into $7/5 < x_n$. Here b is in $\mathcal{C}$, as the difference of (x_n), which is in $\mathcal{C}$ by Theorem 4.6.1, and a constant (Theorem 4.5.3 (a), (b)). From $7/5 < x_n$, $b_n = x_n - 1 > 2/5$ for every n, and $\delta_b = 2/5$, $N_b = 0$ work. The proof then gives the threshold $N = \max\{1, 0\} = 1$, the margin $1/2 + 2/5 = 9/10$ for $a + b$, and the margin $(1/2)(2/5) = 1/5$ for ab.

Exercise. B2

Let $a, b \in \mathcal{C}$. Prove that $[a] < [b]$ if and only if there are $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $b_n - a_n \geq \delta$

for every $n \geq N$.

Solution. By Definition 5.3.2, $[a] < [b]$ means $[b] - [a] \in P$. By Definition 5.2.2, $[b] - [a] = [b - a]$, where $b - a \in \mathcal{C}$ is the termwise difference (Theorem 4.5.3 (b)). An element of $\mathbb{R}$ lies in P if and only if one (equivalently, every) representative of it is positive, by the definition of P and Proposition 5.3.1. So $[a] < [b]$ holds if and only if $b - a$ is positive, which, by Definition 5.3.1, is the condition stated, since the n-th term of $b - a$ is $b_n - a_n$.

Exercise. B3

(a) Let $a \in \mathcal{C}$ with $0 \leq a_n \leq 1$ for every $n \in \mathbb{N}$. Prove that $0_{\mathbb{R}} \leq [a] \leq 1_{\mathbb{R}}$.

(b) Give an example of $a \in \mathcal{C}$ with $0 < a_n < 1$ for every n and $[a] = 1_{\mathbb{R}}$. What does your example show about part (a)?

Solution.

(a) The constant sequences $(0)_n$ and $(1)_n$ are in $\mathcal{C}$ (Theorem 4.5.3 (a)). By Lemma 5.3.4 with $N_0 = 0$, $(0)_n \leq a$ termwise gives $0_{\mathbb{R}} = [(0)_n] \leq [a]$, and $a \leq (1)_n$ termwise gives $[a] \leq [(1)_n] = 1_{\mathbb{R}}$.

(b) Let $a_n := 1 - 1/(n+2)$. For every n, $n + 2 \geq 2 > 1$, so $0 < 1/(n+2) < 1$ (Proposition 3.2.2 (b)), and $0 < a_n < 1$. The sequence $a - (1)_n = (-1/(n+2))$ is null: $(1/(n+2))$ is the shift by one of the null sequence $(1/(n+1))$ (Proposition 4.3.2 (b)), and the negative of a null sequence is null (Theorem 4.4.3 (b)). So $a = (1)_n + (-1/(n+2))_n$ is in $\mathcal{C}$, as the sum of a constant and a null sequence (Theorem 4.5.3 (a), (b)). Since $a - (1)_n$ is null, $a \sim (1)_n$ and $[a] = 1_{\mathbb{R}}$. The example shows that part (a) cannot be sharpened to strict inequalities: $a_n < 1$ for every n, and still $[a] = 1_{\mathbb{R}}$. This is the warning after Lemma 5.3.4 once more. (Any a with $0 < a_n < 1$ for all n and $1 - a_n$ null is a correct answer.)

Exercise. B4: Prove or disprove

Let $a, b \in \mathcal{C}$. Prove each statement or give a counterexample.

(a) If $a_n > 0$ for every n, then $[a] \geq 0_{\mathbb{R}}$.

(b) If a is positive and $b \sim a$, then $b_n > 0$ for every n.

(c) If $a \not\sim b$, then $a - b$ or $b - a$ is positive.

(d) If a is positive and $b_n \geq a_n - 1/(n+1)$ for every n, then b is positive.

(e) If a and b are positive, then $a - b$ or $b - a$ is positive.

Solution.

(a) **True.** The constant $(0)_n$ is in $\mathcal{C}$ (Theorem 4.5.3 (a)), and $0 \leq a_n$ for every n, so Lemma 5.3.4 with $N_0 = 0$ gives $0_{\mathbb{R}} = [(0)_n] \leq [a]$.

(b) **False.** Let $b := (-1, 1, 1, 1, \ldots)$, which agrees with $(1)_n$ from index 1 on. Then $b \in \mathcal{C}$ and $b \sim (1)_n$ (Example (Finite Changes and Shifts) (a)), and $(1)_n$ is positive, with

$\delta = 1$, $N = 0$. But $b_0 = -1$.

(c) **True.** Since $a \not\sim b$, $[a] \neq [b]$ (Sets and Functions, Theorem 5.2.1 (b)), so $[a - b] = [a] - [b] \neq 0_{\mathbb{R}}$. By Theorem 5.3.2 (i), $[a - b] \in P$ or $-[a - b] = [b - a] \in P$. Every representative of an element of P is positive (Proposition 5.3.1), so $a - b$ or $b - a$ is positive.

(d) **True**, margin first. Fix $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $a_n \geq \delta$ for every $n \geq N$, and name the margin $\delta/2 \in \mathbb{Q}_{>0}$ for b. Only then the index: Corollary 3.2.4 gives $K \in \mathbb{Z}_{>0}$ with $1/m < \delta/2$ for every integer $m \geq K$. For $n \geq \max\{N, K\}$, $n + 1 > K$, so

$$b_n \geq a_n - \frac{1}{n+1} > \delta - \frac{\delta}{2} = \frac{\delta}{2}.$$

So b is positive, with margin $\delta/2$ and threshold $\max\{N, K\}$.

(e) **False.** Let $a = b = (1)_n$, both positive. Then $a - b = b - a = (0)_n$, which is not positive: no $\delta \in \mathbb{Q}_{>0}$ has $0 \geq \delta$.

5.3.6.3 C. Going deeper

Exercise. C1, used later

(a) Show that the rule $[a] \mapsto [(|a_n|)_n]$ is a well-defined function $\mathbb{R} \to \mathbb{R}$. Here is the beginning of a proof. *Name the move, then complete it.* "Let $a, a' \in \mathcal{C}$ with $a \sim a'$. The sequences $(|a_n|)$ and $(|a'_n|)$ are in $\mathcal{C}$: by the reverse triangle inequality in $\mathbb{Q}$ (Proposition 4.1.3 (f)), $||a_m| - |a_n|| \leq |a_m - a_n|$, so each has the thresholds of the original. . . ."

(b) Prove that this function is $x \mapsto |x|$: $|[a]| = [(|a_n|)_n]$ for every $a \in \mathcal{C}$.

Solution.

(a) *The move* is **representatives, then independence**: the rule is defined on a representative, so we must show that replacing a by an equivalent a' gives the same output class. Then each class has exactly one output, and the rule is a function (Sets and Functions, Definition 2.1.1).
Completion. Let $e_n := |a_n| - |a'_n|$. By the reverse triangle inequality in $\mathbb{Q}$ (Proposition 4.1.3 (f)),

$$0 \leq |e_n| \leq |a_n - a'_n| \quad \text{for every } n \in \mathbb{N}.$$

The sequence $(a_n - a'_n)$ is null, since $a \sim a'$ (Definition 5.1.2); so is $(|a_n - a'_n|)$, because $||a_n - a'_n| - 0| = |a_n - a'_n - 0|$ and the two sequences pass the same tests of Definition 4.3.3. The squeeze Proposition 4.4.4 (c) in $\mathbb{Q}$, with lower sequence $(0)_n$ and upper sequence $(|a_n - a'_n|)$, shows that $(|e_n|)$ is null. Then (e_n) is null, for the same reason as before: $|e_n - 0| = ||e_n| - 0|$. So $(|a_n|) \sim (|a'_n|)$, and $[(|a_n|)_n] = [(|a'_n|)_n]$ (Sets and Functions, Theorem 5.2.1 (b)).
So the output depends only on the class $[a]$, and the rule defines a function $\mathbb{R} \to \mathbb{R}$.

(b) *Equality.* We use three facts. A class $[b]$ with b eventually equal to b' equals $[b']$ (Example (Finite Changes and Shifts)). A null sequence represents $0_{\mathbb{R}}$ (proof of Theorem 5.3.2 (i)). And, in the ordered field $\mathbb{R}$, $|x| = x$ for $x > 0_{\mathbb{R}}$, $|x| = -x$ for $x < 0_{\mathbb{R}}$, and $|0_{\mathbb{R}}| = 0_{\mathbb{R}}$ (Definition 4.1.1).

Case 1: a is null. Then $[a] = 0_{\mathbb{R}}$, so $|[a]| = 0_{\mathbb{R}}$. The sequence $(|a_n|)$ is null too, since

$||a_n| - 0| = |a_n| = |a_n - 0|$ (Proposition 4.1.3 (a) and Definition 4.1.1), so the two sequences pass the same tests of Definition 4.3.3. Hence $[(|a_n|)_n] = 0_{\mathbb{R}} = |[a]|$.

Case 2: a is not null. By Theorem 4.5.4 with $F = \mathbb{Q}$, fix $c \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $a_n \geq c$ for every $n \geq N$, or with $a_n \leq -c$ for every $n \geq N$.

If $a_n \geq c$ for $n \geq N$, then a is positive, so $[a] \in P$ and $[a] > 0_{\mathbb{R}}$, whence $|[a]| = [a]$. For $n \geq N$, $a_n > 0$, so $|a_n| = a_n$. The sequences $(|a_n|)$ and a agree from N on, so $[(|a_n|)_n] = [a] = |[a]|$.

If $a_n \leq -c$ for $n \geq N$, then $-a$ is positive, so $-[a] = [-a] \in P$ and $[a] < 0_{\mathbb{R}}$, whence $|[a]| = -[a] = [-a]$. For $n \geq N$, $a_n < 0$, so $|a_n| = -a_n$. The sequences $(|a_n|)$ and $-a$ agree from N on, so $[(|a_n|)_n] = [-a] = |[a]|$.

Part (b) does **not** say that $|a_n| = a_n$ or $-a_n$ for all n; a representative of a positive number may have negative terms early on, as the late-positivity example shows. It says that the class of $(|a_n|)$ is the right one, and classes do not see early terms.

Marking scheme. Part (a), 10 points. - 2: names the move (representatives, then independence) and states the goal $(|a_n|) \sim (|a'_n|)$. - 3: proves $||a_n| - |a'_n|| \leq |a_n - a'_n|$ for every n, citing the reverse triangle inequality in $\mathbb{Q}$. - 3: from $a \sim a'$ and the inequality of the previous step, shows that $(|a_n| - |a'_n|)_n$ is null. - 2: concludes $[(|a_n|)_n] = [(|a'_n|)_n]$, so each class has one output and the rule is a function $\mathbb{R} \to \mathbb{R}$.

A direct ε–N proof of the third step, with no squeeze, earns its 3 points: given $\varepsilon \in \mathbb{Q}_{>0}$, take N for the null sequence $(a_n - a'_n)$; then $||a_n| - |a'_n|| \leq |a_n - a'_n| < \varepsilon$ for every $n \geq N$. Fatal errors (0 for the step): using $|x|$ on $\mathbb{R}$, or citing part (b); proving only that the two output sequences are Cauchy, without proving them equivalent. Part (b) is not marked.

Exercise. C2: Find the error

Here is an argument for Proposition 5.3.1.

"Let a be positive and $a' \sim a$. Fix $\delta \in \mathbb{Q}_{>0}$ and N with $a_n \geq \delta$ for every $n \geq N$. Since $a' - a$ is null, there is N' with $|a'_n - a_n| < \delta$ for every $n \geq N'$. So, for $n \geq \max\{N, N'\}$, $a'_n > a_n - \delta \geq 0$. Hence $a'_n > 0$ for all large n. Therefore a' is positive."

Is the conclusion true? Does the argument prove it? If not, name the line that fails, say why, give a counterexample to that step, and repair the argument.

Solution (Hint 1). Write out in symbols what "a' is positive" requires, and compare it with what the argument has when it reaches the last line.

Solution (Hint 2). "Positive" asks for one margin $\delta' \in \mathbb{Q}_{>0}$, chosen before n, with $a'_n \geq \delta'$ for all large n. Find the line where the argument should produce such a δ', and ask what it produces instead.

Solution. The conclusion is true: it is Proposition 5.3.1.

The middle lines are correct. With the tolerance δ and the strict inequality $|a'_n - a_n| < \delta$,

we get $a'_n > a_n - \delta \geq \delta - \delta = 0$ for every $n \geq \max\{N, N'\}$. So the line "$a'_n > 0$ for all large n" is proved.

The failing line is the last: "Therefore a' is positive." What has been proved is

$$\exists N \in \mathbb{N} \;\; \forall n \geq N \;\; a'_n > 0,$$

while "positive" (Definition 5.3.1) is

$$\exists \delta \in \mathbb{Q}_{>0} \;\; \exists N \in \mathbb{N} \;\; \forall n \geq N \;\; a'_n \geq \delta.$$

To pass from the first to the second, one would take the margin $\delta := a'_n$, which depends on n, and then claim one margin for all n. That swaps a "for every n, there is a margin" for a "there is a margin for every n". It is the analogue, for a margin depending on n, of entry (N2) of Logic and Proof, Example (What is not allowed), where a threshold depending on ε is treated as uniform.

Counterexample to the step. The sequence $(1/(n+1))$ has $a'_n > 0$ for all n, so it satisfies the premise of the last line, and it is not positive (the non-example after Definition 5.3.1). So "eventually > 0" does not imply "positive". The argument reached a true conclusion through a step that is not valid.

Repair. Use the tolerance $\delta/2 \in \mathbb{Q}_{>0}$ (Proposition 4.1.2 (b)) instead of δ. Since $a' - a$ is null, choose N'' with $|a'_n - a_n| < \delta/2$ for every $n \geq N''$. For $n \geq \max\{N, N''\}$,

$$a'_n > a_n - \frac{\delta}{2} \geq \delta - \frac{\delta}{2} = \frac{\delta}{2}.$$

So $a'_n \geq \delta/2$ for every $n \geq \max\{N, N''\}$: one margin, $\delta/2$, fixed before n, and a' is positive. This is the proof of Proposition 5.3.1.

Marking scheme. (10 points) - 2: says the conclusion is true, citing Proposition 5.3.1. - 3: identifies the last line as the failing one and names the quantifier swap (a margin depending on n treated as one margin for all n). - 2: a counterexample to the step: a sequence whose terms are eventually > 0 and which is not positive, such as $(1/(n+1))$. - 3: a repair that produces one margin in $\mathbb{Q}_{>0}$ fixed before n (for instance the tolerance $\delta/2$ above; any tolerance strictly between 0 and δ works equally well). Fatal errors (0 for the step): blaming one of the valid middle lines; a repair whose margin still depends on n.